

Comparative Efficacy and Safety of Monoclonal Antibodies for Alzheimer’s Disease

A Bayesian Network Meta-Analysis — Project Report

Wandering Data Ghost #2

Overview

This project conducted a systematic review and Bayesian network meta-analysis comparing the efficacy and safety of 14 monoclonal antibodies for Alzheimer’s disease across 36 randomized controlled trials (22,670 patients). The complete analytical pipeline was executed—from database searching and dual-reviewer screening to risk-of-bias assessment, MCMC network model fitting, SUCRA-based treatment ranking, and inconsistency testing—using R 4.3.2 and STATA 15.0.

The manuscript was not submitted for publication: the analysis was completed during the undergraduate-to-graduate application period, and by the time I returned to write it up, a closely related paper had already appeared in the literature. With the exception of the back-to-back literature screening (conducted with a second reviewer as per systematic review protocol), the full study design, data extraction, and statistical analysis were carried out independently by the author.

Background

Alzheimer’s disease (AD) is the leading cause of dementia worldwide, imposing a substantial burden on patients, caregivers, and healthcare systems. Despite decades of research, disease-modifying therapy remains elusive. The accumulation of amyloid- β plaques and tau neurofibrillary tangles are the two principal pathological hallmarks of AD, and monoclonal antibodies targeting these proteins have emerged as a promising therapeutic strategy—with multiple agents reaching Phase II and Phase III trials.

However, individual trials have evaluated different antibodies, doses, and patient populations, and direct head-to-head comparisons are sparse. Previous pairwise meta-analyses cannot simultaneously rank competing interventions. A Bayesian network meta-analysis (NMA) overcomes this limitation by integrating direct and indirect evidence, enabling probabilistic ranking of all treatments on both efficacy and safety outcomes.

Methods

Search and Eligibility

A systematic search was conducted in PubMed, Embase, Cochrane Library, and Web of Science (up to September 18, 2023) using combinations of MeSH terms and free-text keywords for AD and monoclonal antibodies (detailed strategy in Supplementary 1). Eligible studies were randomized controlled trials enrolling adults meeting AD diagnostic criteria, evaluating a monoclonal antibody versus placebo, and reporting at least one prespecified outcome. Efficacy outcomes: ADAS-Cog (lower = better), CDR-SB (lower = better), MMSE (higher = better). Safety outcomes: adverse events (AEs), serious adverse events (SAEs), and deaths. Two reviewers independently screened all records; disagreements were resolved by discussion or a third reviewer.

Exclusion criteria: duplicate publications, animal studies, case reports, conference abstracts, reviews, unavailable full texts, and studies in populations with comorbid major organic diseases.

Data Extraction and Risk-of-Bias Assessment

Extracted fields included first author, publication year, country, sample size, sex, age, intervention and dose, comparator, and outcomes. Methodological quality was assessed with the Cochrane Risk of Bias 2.0 (RoB2) tool across five domains: randomization process, deviations from intended interventions, missing outcome data, outcome measurement, and selective reporting. Each domain was rated low risk, some concerns, or high risk.

Statistical Analysis

A Bayesian random-effects NMA was performed in R 4.3.2 using diffuse-prior random-effects models; multi-arm trials were handled within the same framework. Posterior estimates were obtained via Markov chain Monte Carlo (MCMC) sampling; convergence was verified using trace plots and Brooks–Gelman–Rubin diagnostics. Continuous outcomes (ADAS-Cog, CDR-SB, MMSE) were summarized as mean differences (MD) with 95% credible intervals (CrI); binary safety outcomes as odds ratios (OR) with 95% CrI. The surface under the cumulative ranking curve (SUCRA) was computed to estimate each treatment’s probability of being best. Global inconsistency was assessed using the unrelated mean effects (UME) model; local inconsistency was tested via node-splitting for outcomes with closed loops. Network plots and comparison-adjusted funnel plots were produced in STATA 15.0; forest plots, cumulative ranking plots, rank-probability plots, trace plots, and Brooks–Gelman–Rubin diagnostics were exported from R.

Results

Study Selection

The initial search retrieved 2,740 records. After removing 601 duplicates and 949 non-RCT records, 971 were excluded on title/abstract screening and 192 on full-text review, leaving **36 RCTs** (22,670 patients) for the final analysis. Included drugs: GSK933776, bapineuzumab, aducanumab, crenezumab, AAB-003, solanezumab, CAD-106, tilavonemab, ponezumab, donanemab, lecanemab, gantenerumab, BIIB092, and semorinemab.

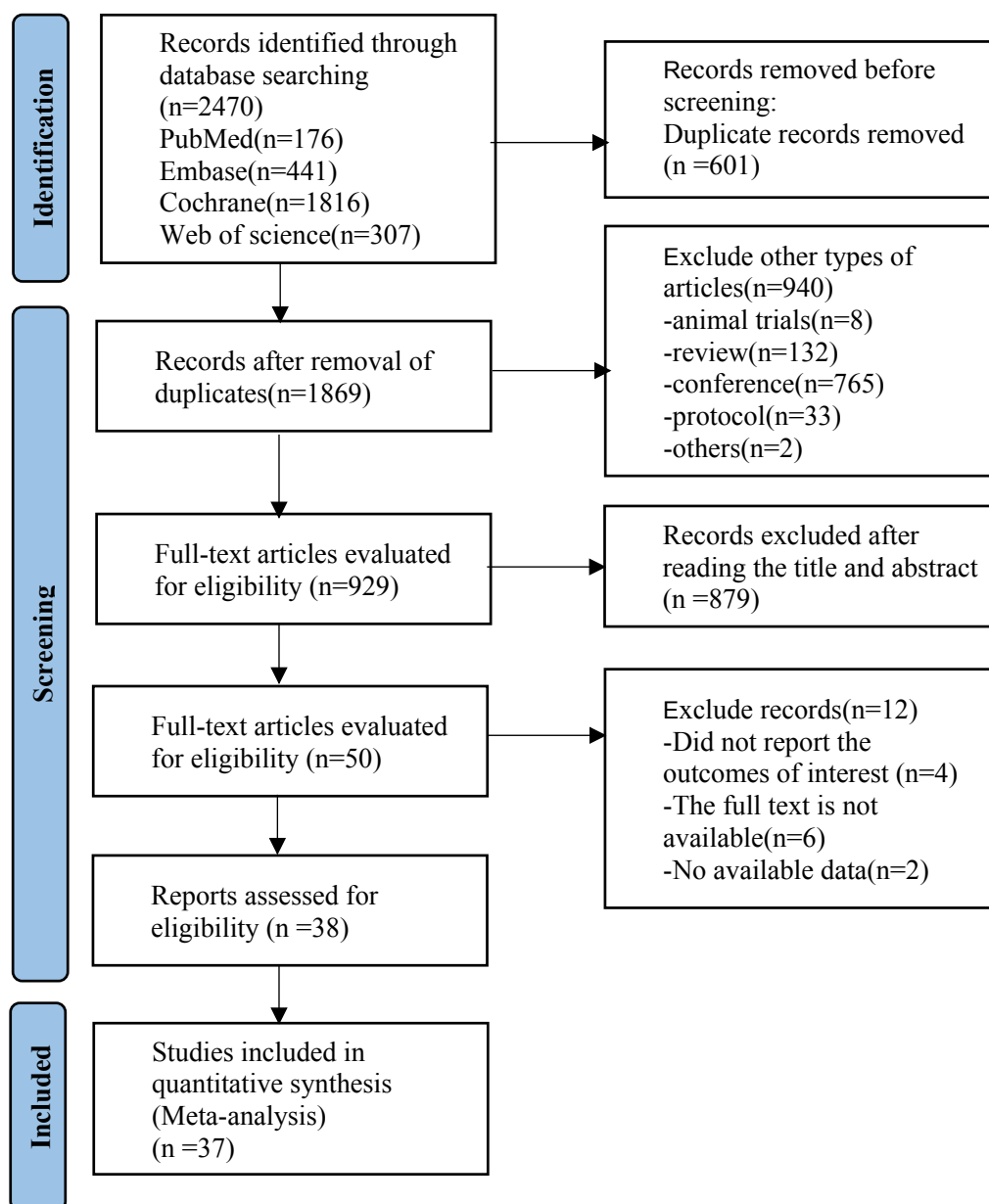


Figure 1: PRISMA flow diagram of study selection.

Risk of Bias and Evidence Networks

All included studies reported their blinding procedures clearly. The main source of high risk arose from deviations from intended interventions. The current scripted export set includes evidence networks for ADAS-Cog and AEs; node size is proportional to participant numbers and edge thickness to the number of direct comparisons.

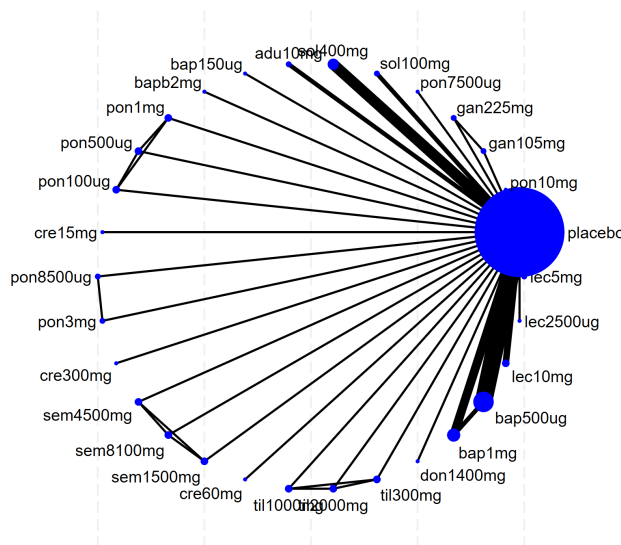


Figure 2: Evidence network for ADAS-Cog. Node size $\propto$ participants; edge width $\propto$ number of direct comparisons.

Efficacy Outcomes

ADAS-Cog (20 studies; lower = better). Closed loops were present; local inconsistency testing showed no difference between direct, indirect, and network comparisons for bapineuzumab 500 μ g vs. 1 mg. Compared with placebo, three tilavonemab doses significantly reduced ADAS-Cog: til 1000 mg (MD = -9.0 , 95% CrI -12.0 to -5.8), til 300 mg (MD = -8.8 , 95% CrI -12.0 to -5.5), and til 2000 mg (MD = -7.4 , 95% CrI -11.0 to -4.0). SUCRA ranking: til 1000 mg > til 300 mg; pon 10 mg ranked lowest.

CDR-SB (12 studies; lower = better). Lecanemab 10 mg was the only intervention significantly superior to placebo (MD = -0.40 , 95% CrI -0.75 to -0.037). SUCRA ranking: lec 10 mg highest; bap 1 mg lowest.

MMSE (12 studies; higher = better). Closed loops were present; local inconsistency testing again showed no meaningful inconsistency for bapineuzumab comparisons. Tilavonemab significantly improved MMSE vs. placebo: til 2000 mg (MD = $+4.3$, 95% CrI 2.5 to 6.1), til 300 mg (MD = $+3.9$, 95% CrI 2.1 to 5.7), til 1000 mg (MD = $+3.2$, 95% CrI 1.4 to 5.0).

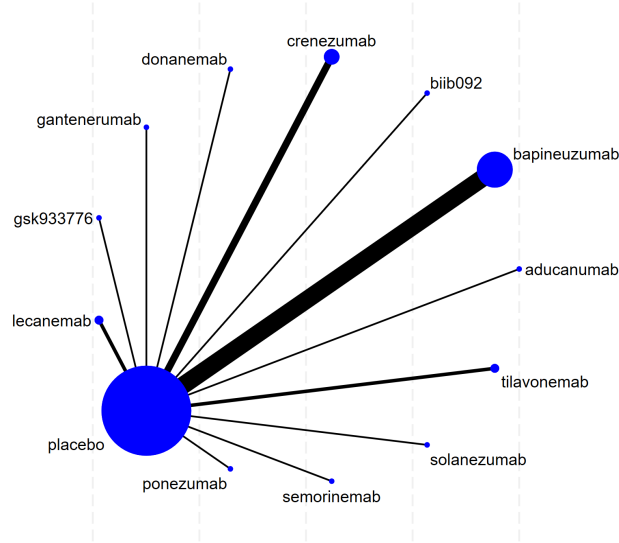


Figure 3: Evidence network for adverse events (AEs). Node size $\propto$ participants; edge width $\propto$ number of direct comparisons.

SUCRA ranking: til 2000 mg > til 300 mg > til 1000 mg; bap 40 mg ranked lowest.

The current R export set for ADAS-Cog includes the forest plot, cumulative ranking (SUCRA) plot, Brooks–Gelman–Rubin diagnostic, and rank-probability plot shown below.

Safety Outcomes

Adverse events (26 studies). Compared with placebo, lecanemab was associated with significantly fewer AEs (OR = 0.49, 95% CrI 0.31–0.74). SUCRA ranking: ponezumab highest (fewest AEs), aducanumab second; GSK933776 lowest.

Serious adverse events (20 studies). Ponezumab was associated with significantly more SAEs than placebo (OR = 7.7, 95% CrI 3.8–20.7). SUCRA ranking: solanezumab highest (fewest SAEs); ponezumab lowest.

Deaths (11 studies). No intervention showed a statistically significant difference in mortality vs. placebo. SUCRA ranking: lecanemab highest; crenezumab lowest.

The current R export set for AEs includes the forest plot, Brooks–Gelman–Rubin diagnostic, and rank-probability plot shown below.

Summary of Comparative Results

* Only statistically significant result vs. placebo for AEs; SUCRA top-ranked intervention is ponezumab.

† Ponezumab was associated with significantly more SAEs; solanezumab ranked best (fewest SAEs) by SUCRA.

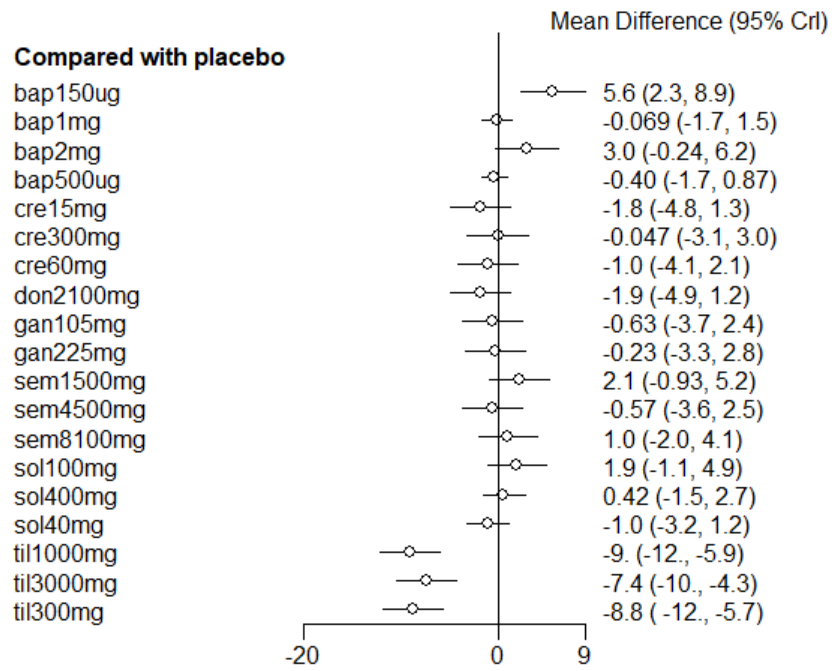


Figure 4: Network forest plot: all interventions vs. placebo for ADAS-Cog.

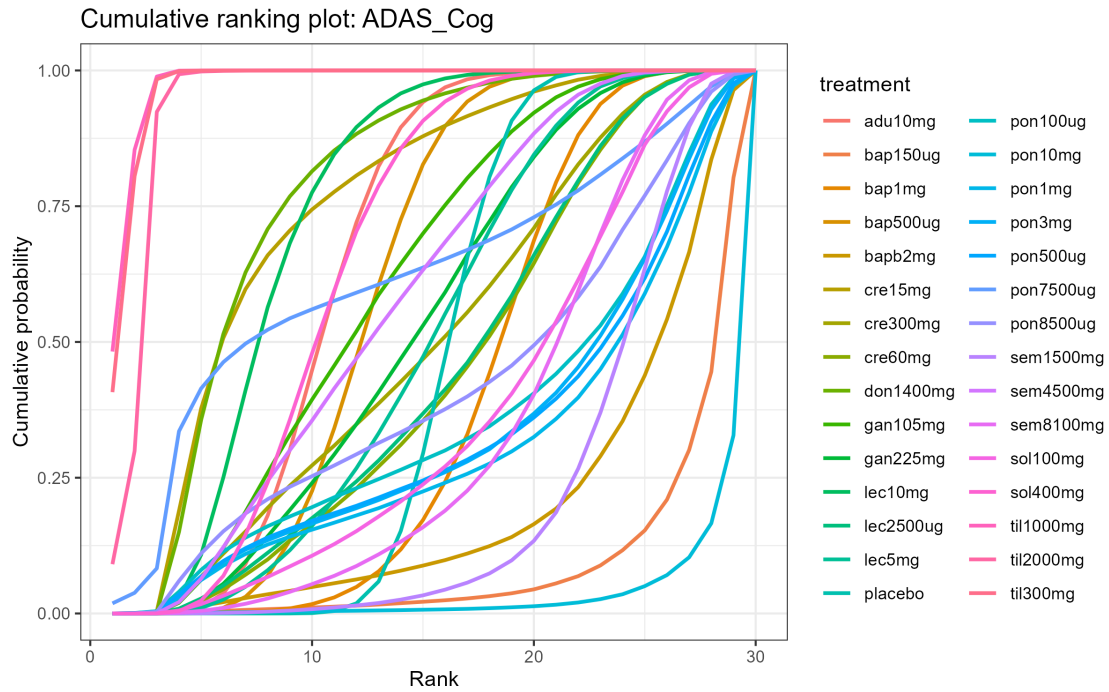


Figure 5: Cumulative ranking (SUCRA) plot for ADAS-Cog.

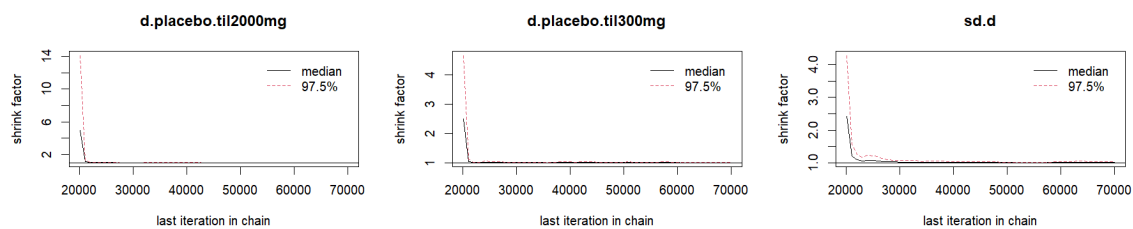


Figure 6: Brooks–Gelman–Rubin convergence diagnostic for the ADAS-Cog model.

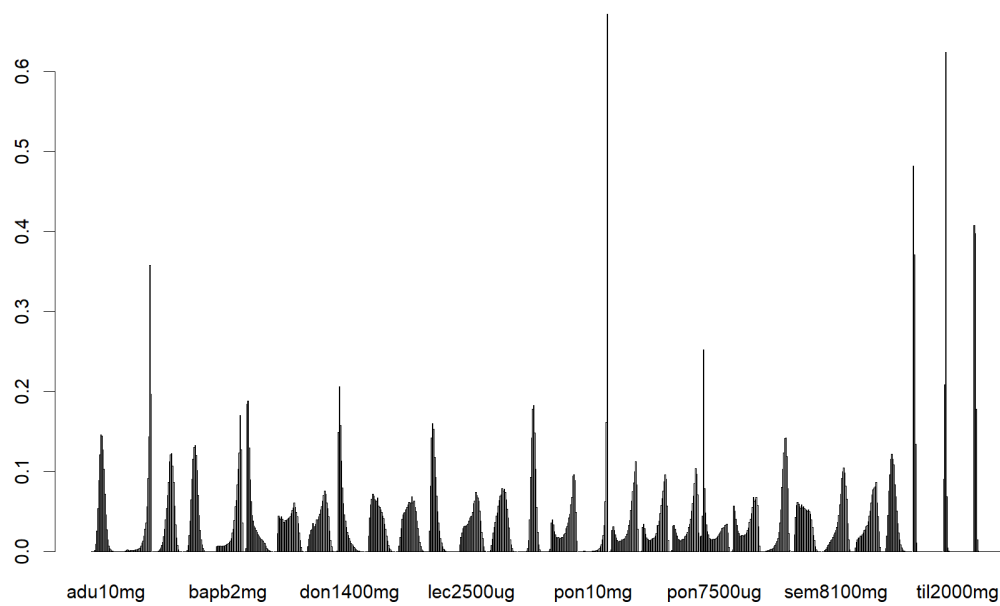


Figure 7: Rank-probability plot for ADAS-Cog.

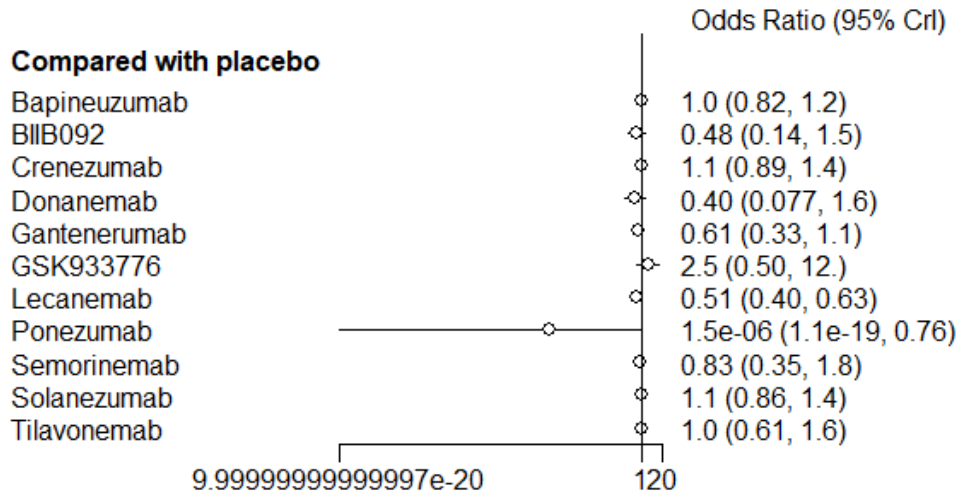


Figure 8: Network forest plot: all interventions vs. placebo for adverse events (AEs).

Table 1: Summary of Bayesian NMA results (estimates vs. placebo; SUCRA reflects probability of being best-ranked).

Outcome	Top-ranked intervention		Effect estimate (95% CrI)	Direction
ADAS-Cog	tilavonemab		MD = −9.0 (−12.0, −5.8)	lower = better
CDR-SB	lecanemab 10 mg		MD = −0.40 (−0.75, −0.037)	lower = better
MMSE	tilavonemab 2000 mg		MD = +4.3 (+2.5, +6.1)	higher = better
AEs	ponezumab (SUCRA)	(SUCRA)	lecanemab: OR = 0.49 (0.31, 0.74)*	lower OR = safer
SAEs	solanezumab (SUCRA)	(SUCRA)	ponezumab: OR = 7.7 (3.8, 20.7)†	lower OR = safer
Deaths	lecanemab (SUCRA)	(SUCRA)	no significant difference	—

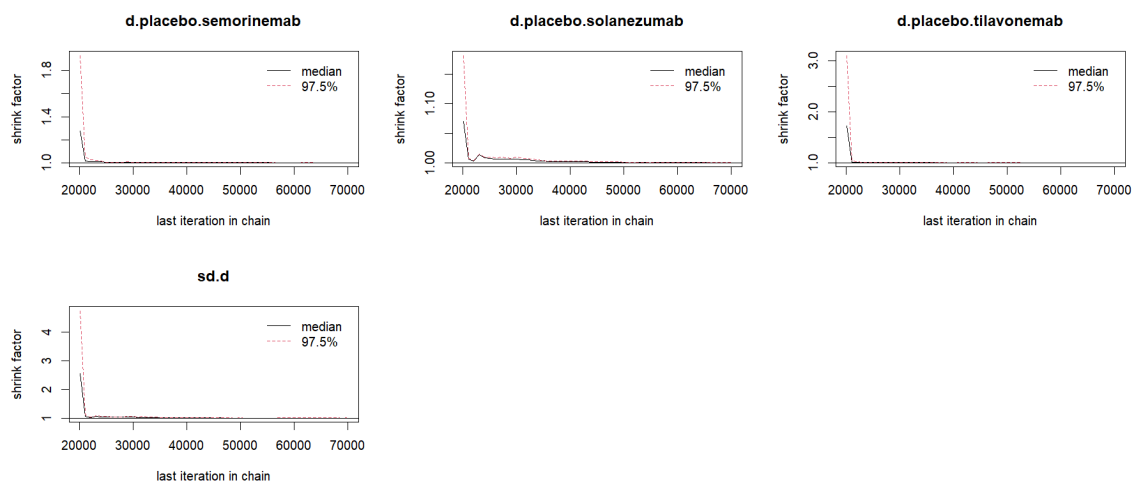


Figure 9: Brooks–Gelman–Rubin convergence diagnostic for the adverse-events model.

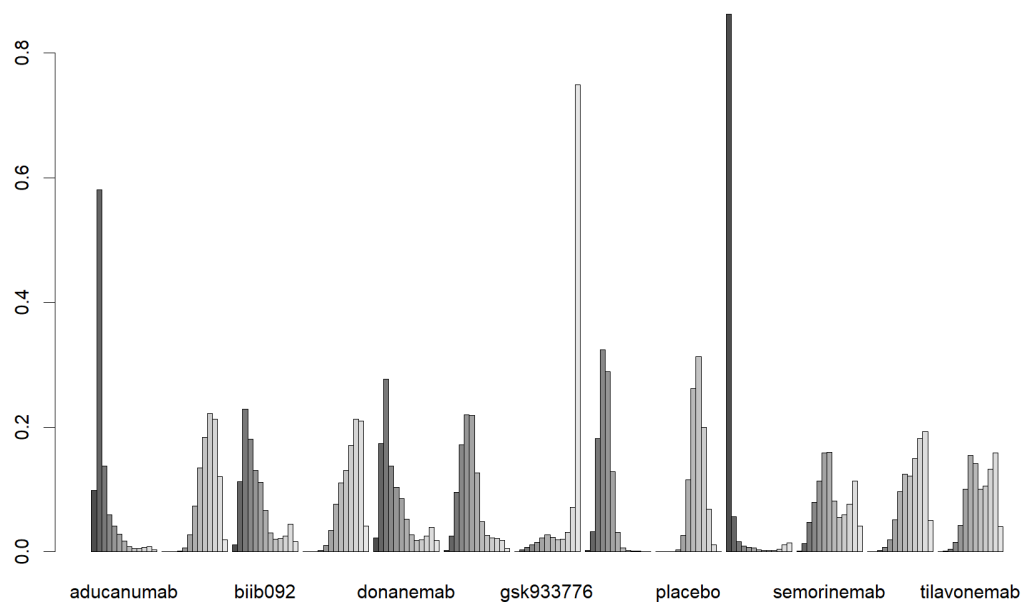
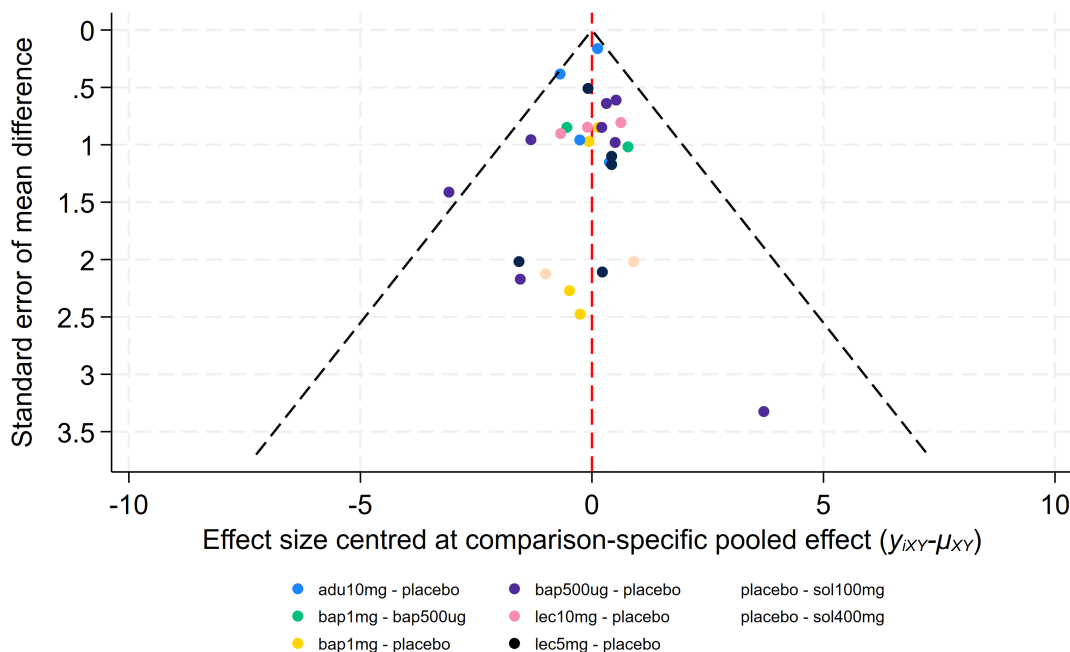


Figure 10: Rank-probability plot for adverse events (AEs).

Publication Bias and Consistency

Comparison-adjusted funnel plots were exported for ADAS-Cog and AEs. In the currently reproduced workflow, node-splitting output was available for ADAS-Cog and did not indicate important disagreement between direct and indirect estimates.



Note: Comparisons including only one study (when present) have been excluded

Figure 11: Comparison-adjusted funnel plot for ADAS-Cog.

Discussion

Monoclonal antibodies targeting amyloid- β or tau remain a major focus of disease-modifying therapy for AD. Prior pairwise meta-analyses have been limited to single-drug comparisons. This NMA synthesizes 36 trials across 14 drug compounds and six outcomes, providing a unified efficacy and safety profile for each drug and dose.

The principal finding is that tilavonemab, a tau-targeting antibody, consistently ranked among the most effective interventions on both cognitive scales (ADAS-Cog and MMSE), suggesting that tau pathology may be a particularly tractable target in the evaluated study populations. Lecanemab, an amyloid-targeting antibody, was the only agent to show a statistically significant improvement on CDR-SB and simultaneously demonstrated a favorable safety profile (fewer AEs vs. placebo), aligning with its subsequent FDA approval.

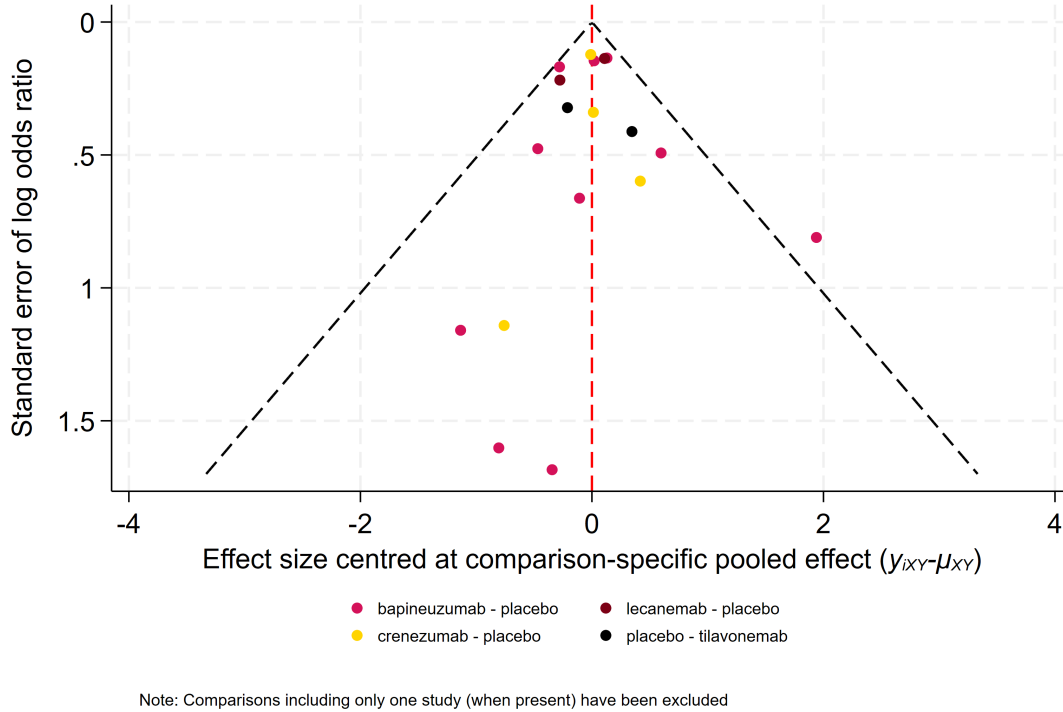


Figure 12: Comparison-adjusted funnel plot for adverse events (AEs).

Ponezumab, by contrast, was associated with a markedly elevated SAE risk, underscoring the importance of joint efficacy and safety evaluation rather than relying on a single outcome.

Limitations

1. *Scale heterogeneity:* ADAS-Cog, CDR-SB, and MMSE differ in sensitivity, administration, and clinical anchoring; cross-drug comparisons on a single scale may not translate uniformly to clinical significance.
2. *Follow-up heterogeneity:* Outcome assessment time points varied across trials, potentially introducing heterogeneity in pooled estimates.
3. *Population heterogeneity:* Included studies differed in disease stage, biomarker confirmation, and background therapy, which may limit the generalizability of ranking results.

Code Availability

The cleaned input data, the standalone R workflow, the STATA do-file, and the current figure-generation pipeline are publicly available in the following GitHub repository.